DO NOW — RECALL

Lesson 078 · Binary Search: Bounds and Sorted Data

Name: _____ Class: _____ Date: _____

1. Binary search precondition?

2. What does low represent?

3. What is mid for bounds low0 high6?

4. Binary search target15 in [1,3,5,7,9,11,13,15]: give midpoint indexes.

DO NOW — CHECK / OBJECTIVES

Correct and improve

Check page 1 in GREEN PEN. Write corrections in full.

1. Model answer

Sorted data.

2. Model answer

First possible candidate index.

3. Model answer

Index3.

4. Model answer

mid3=7, mid5=11, mid6=13, mid7=15; found index7.

By the end, you can…

• Use binary search only when elements are sorted; keep inclusive low and high candidate bounds.

• At every mismatch discard the midpoint and half the candidates with low=mid+1 or high=mid−1.

• Each unsuccessful step shrinks the search interval. Worst time O(log n), iterative extra memory O(1).

Vocabulary — know these terms

low — Lowest candidate index.

high — Highest candidate index.

midpoint — Middle candidate.

halving — Repeatedly reducing interval by about half.

NEW LEARNING

Read, annotate and explain

Understand

Use binary search only when elements are sorted; keep inclusive low and high candidate bounds.

Trace

At every mismatch discard the midpoint and half the candidates with low=mid+1 or high=mid−1.

Explain

Each unsuccessful step shrinks the search interval. Worst time O(log n), iterative extra memory O(1).

Explain one step in your own words, using the correct technical terms.

I DO — WORKED EXAMPLE

Follow the worked method

Find14 in [2,5,8,11,14,17,20].

Check the working

mid3=11, mid5=17, mid4=14, returns index4.

WE DO — GUIDED PRACTICE

Try → Discuss → Improve

YOU DO — INDEPENDENT PRACTICE

Questions · Part 1

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

Question 1

Trace target11 in [1,3,5,7,9,11,13]. [6]

Question 2

Write three binary-search comparison cases and correct bounds. [6]

YOU DO — INDEPENDENT PRACTICE

Questions · Part 2

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

Question 3

Compare iterative binary and linear search with respect to time and precondition. [6]

EXIT TICKET / REFLECTION

Check your understanding

Complete the actual lesson Exit Ticket before checking the answer.

After a too-small midpoint, setting low=mid always ensures progress.

How confident are you? Circle one: Confident / Getting there / Need help

Explain your choice, identifying one concept you can now use independently.

GREEN PEN — Correct and explain

Choose one mistake from your independent work. Write the correct method and explain precisely why your original answer was wrong.

Next step: revisit the model or a relevant revision-guide section if needed.